Energy 309 (2024) 133088

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Design and optimal scheduling of forecasting-based campus multi-energy complementary energy system

Weichao Dong [a,*], Hexu Sun [a], Zheng Li [a], Huifang Yang [b]

[a] School of Electrical Engineering, Hebei University of Science and Technology, Shijiazhuang, 050018, China
[b] Academic Affairs Office, Shijiazhuang Tiedao University, Shijiazhuang, 050043, China

ARTICLE INFO

Keywords:
Campus multi-energy complementary energy system
Hybrid forecasting model
Multi-energy complementary energy system model
Multi-objective optimal model

ABSTRACT

This study presents a complete campus multi-energy complementary energy system (MCES), including an accurate forecasting model, efficient MCES model, and effective multi-objective optimal scheduling strategy to better utilize renewable energy. A hybrid forecasting model, including multi-scale mathematical morphological decomposition, a bidirectional long short-term memory network, and subsequences to the original sequence (S2O) manner based on the rolling approach (RA), is utilized to forecast renewable energy variations. RA continuously updates input datasets to improve forecasting accuracy. Decomposition and forecasting modules are employed in an S2O manner to reduce the number of required modules and forecasting cost. The volatility of renewable energy is mitigated by supplementing energy sources with storage. During operation, the conversion times of different energies are reduced by reasonably planning the energy supply sequence based on different loads on the demand side, increasing the energy utilization rate. The proposed multi-objective optimal scheduling strategy includes a stacked multilevel-denoising autoencoder, non-dominated sorting genetic algorithm-II, and deep reinforcement learning (DRL) for surrogate-model building, Pareto frontier establishment, and optimal solution selection. This is the first study to use DRL to select the final optimal solution. A performance comparison confirms the proposed model effectively decreases costs and pollution while increasing thermal comfort.

1. Introduction

Reducing carbon emissions and environmental pollution is a global environmental concern. However, the demand for energy keeps increasing, making it a considerable challenge for the energy sector to supply adequate energy while reducing greenhouse emissions [1]. Renewable energy, considered the most effective solution to the energy problem, is highly volatile and intermittent. Thus, it cannot effectively replace traditional fossil fuels as the main source of energy supply [2]. A multi-energy complementary energy system (MCES) is an integrated system that involves energy generation, transmission, storage, and consumption. It is considered a novel means to effectively utilize renewable energy, owing to its low emissions and high energy efficiency [3,4].

Numerous studies have been conducted on MCES. For example, an MCES that considers thermal integration and process synergy has been proposed [5]. The ε-constraint method, combined with a technique for order preference similar to an ideal solution (TOPSIS), was used to establish the optimal scheduling model and realize multi-objective optimization. Zhang et al. [6] designed a novel superstructure for an MCES and established two-stage stochastic programming using mixed-integer linear programming, considering multiple uncertainties to minimize the annual cost. Yang et al. [7] emphasized the integrated demand-side management of an MCES via multi-energy pricing and non-cooperative games. Feng et al. [8] proposed a multi-objective optimal scheduling model for MCES using the information gap decision theory to analyze the effect of uncertainties on the demand and supply sides. An MCES comprising integrated photovoltaic (PV)–thermal technology was established for a near-zero-energy building, and the load characteristics and energy-saving rate were analyzed [9]. Wang et al. [10] reviewed multi-energy complementary energy systems based on solar energy, providing many novel models of MCES. Lu et al. [11] developed a community MCES and proposed a multi-criteria compromise ranking decision-making approach to complete the scheduling process. Han et al. [12] established a solar-based integrated energy system and proposed a multi-objective sustainability optimization method. The district MCES and its hierarchical distributed optimal scheduling strategy are established in Ref. [13].

* Corresponding author.
E-mail address: dongweichao@hebust.edu.cn (W. Dong).

https://doi.org/10.1016/j.energy.2024.133088
Received 28 May 2024; Received in revised form 1 August 2024; Accepted 2 September 2024
Available online 3 September 2024

Although these studies are beneficial, most of them solely focus on one of the aspects of the MCES, such as system modeling and optimal scheduling strategy research. A complete MCES should contain three parts, including an accurate forecasting model, an efficient MCES model, and an effective scheduling strategy. To date, only a few studies have established such a system.

1.1. Forecasting model

Accurate forecasting is essential for integrating renewable energy sources. Based on forecasting results, a scheduling strategy can effectively address problems due to high fluctuations in renewable energy [14]. Several advanced forecasting models have been proposed for efficient forecasting.

Huang et al. [15] proposed a wind speed forecasting model that included a genetic algorithm (GA) and a long short-term memory (LSTM) neural network. The LSTM structure and the corresponding hyperparameters were evolved using a GA, and the results indicated the satisfactory performance of the proposed model. Zheng et al. [16] developed a combinatorial deep learning model comprising a convolutional LSTM and an LSTM and verified its superiority by comparing different benchmark models under three criteria. Hu et al. proposed an advanced hybrid forecasting model, including variable mode decomposition (VMD) and a sparse autoencoder (SAE), a high-order fuzzy cognitive mapping neural network (HFCMNN), and a batch gradient descent optimization (BGDO) algorithm [17]. Two-phase input data preprocessing was performed using VMD and SAE, HFCMNN was used as the core forecasting module, and BGDO optimized the weights and biases. Shang et al. [18] established a novel wind-speed forecasting model. Specifically, outliers and noise were removed via ensemble empirical mode decomposition. Subsequently, the attention mechanism and convolutional neural network (CNN) collectively worked to perform forecasting. Ahn and Hur proposed a practical wind-power forecasting model based on wavelet transform (WT) and an autoregressive integrated moving average with an explanatory variable [19], whose performance was verified by a case study.

An increasing number of advanced models have been developed to complete forecasting tasks. However, most existing forecasting tasks use fixed-coefficient models, indicating that the model is trained using historical data rather than real-time data, thereby greatly affecting the forecasting accuracy. The forecasting accuracy can be improved using a rolling approach (RA) to continuously add real-time data and update input datasets [20]. More specifically, the RA adds new data, removes old data, and keeps the data size unchanged. Simultaneously, most forecasting tasks use hybrid models, which comprise data preprocessing and core forecasting modules. The preprocessing module processes the input data, and the core forecasting module completes the forecasting [21]. An RA-based hybrid forecasting model can yield more accurate forecasting results.

The most crucial preprocessing step is decomposing the original input data into multiple subsequences with different frequencies to improve the forecasting performance. Hu et al. [22] used the rolling VMD method to decompose wind data and obtained accurate forecasting results. However, the VMD and WT decomposition processes must be completed in the frequency domain; that is, the input data must first be converted to the frequency domain and then returned to the time domain after decomposition, which increases the conversion time, costs, and the likelihood of errors. By contrast, multi-scale mathematical morphological decomposition (MMMD) can be directly utilized in the time domain, effectively reducing the conversion time and errors [23]. Thus, using RA-based MMMD can efficiently complete the decomposition process and reduce data leakage [24,25].

Core forecasting modules are developed using deep neural network (DNN) models. Common DNNs include convolutional neural networks (CNNs), stacked denoising autoencoders, and LSTM [26]. The input data depend on previous and subsequent time points, and a bidirectional LSTM (BiLSTM) network can reserve long-term memories in both time orders, effectively improving the forecasting performance [27]. Therefore, using RA-based BiLSTM as the core forecasting module can considerably improve forecasting accuracy.

Although the RA can improve the performance of the forecasting model, a disadvantage remains. The hybrid model that employs the RA needs to repeatedly use multiple modules in the training and testing processes. For example, in testing process, the input data are decomposed into multiple subsequences in the first forecasting step, which are then forecasted. Subsequently, the prediction result of each subsequence is aggregated to obtain the final forecasting result. The process is repeated in a rolling manner until the prediction is complete. The required number of decomposition and forecasting modules is equal to the number of subsequences, and the forecasting of this process incurs high costs. This can be resolved using subsequences to the original sequence (S2O). The decomposed subsequences serve as the input to the S2O method, while the original sequence constitutes the output. The S2O method only requires one module for the decomposition and forecasting steps [28].

Therefore, the hybrid forecasting model, including MMMD, BiLSTM, and S2O based on RA, can achieve improved performance. MMMD can decompose input data directly in the time domain, reducing forecasting expenses and errors. BiLSTM can reserve long-term memories in both time orders, improving the forecasting performance. The S2O method only trains one module for both the decomposition and forecasting steps, thereby reducing the number of required modules and addressing the main disadvantage of RA.

1.2. Modeling of MCES

Different types of MCES have been established to better combine renewable and traditional energy sources. Wang et al. [29] proposed an MCES comprising heating, cooling, and power systems, where a combination of solar energy and gas was used to supply electricity and heat while hydrogen and an energy storage system were used to store the excess energy. Liu et al. [30] introduced a comprehensive MCES method. On the supply side, the system comprised biomass, water, fossil fuels, geothermal, wind, and solar energy, while the demand side comprised heating, cooling, hot water, and electrical loads. However, the established MCES was complex and unsuitable for application in practical scenarios. Guan et al. [31] proposed an MCES that used solar energy as a renewable energy source, along with natural gas, to complete the energy supply. Simultaneously, the system was connected to the power grid to transmit excess electric power.

Although the above studies established different MCES models, several problems remain unaddressed. First, owing to the complexity of the modeling process, most studies have used only one type of renewable energy. However, owing to the intermittent nature of renewable energy, the time period for effective utilization is extremely limited. For instance, solar energy can be used only during the day, and its output is almost zero at night. On the other hand, the output of wind energy is small during the day but increases at night [32]. Thus, combining various renewable energy sources can increase the utilization time. Simultaneously, owing to the volatility of renewable energy, the fluctuation range of the output power is large (between 20 % and 100 %). The high volatility leads to frequent curtailment of wind and solar energies, because of which other energy sources and energy storage systems must be reasonably supplemented to effectively fulfill the energy supply while minimizing emissions. Finally, the demand side typically includes multiple load types. Therefore, it is arduous to complete effective planning to reduce the conversion of different energy types during operation and increase the energy utilization rate.

1.3. Optimal scheduling strategy

The most significant aspect of the MCES is the formulation of an

Table 1
Table of previous works.

Ref.	Forecasting Model	System Modeling	Scheduling Strategy
[5,9,29–31]		✓	
[6,11–13]		✓	✓
[7,8,33–38]			✓
[15–19,22]	✓		
[40,42]	✓		✓

effective, optimal scheduling strategy. Different types of energy inputs and loads exist on the supply and demand sides, respectively. During operation, balancing this highly complex, non-linear, and non-convex supply–demand relationship is crucial, and approaches to reduce operating costs, minimize emissions, and enhance user comfort must also be considered [33]. Therefore, multi-objective optimization is the optimal scheduling process of the MCES, and establishing a multi-objective optimal scheduling strategy (MOSS) to realize effective scheduling of the MCES is challenging.

Chung et al. [34] developed a multi-objective optimization model for a desiccant wheel. Based on the decision variables and metamodel, the non-dominated sorting genetic algorithm-II (NSGA-II) was used to generate the Pareto solution set, and the optimal point was selected via multi-criteria decision-making (MCDM). Bai et al. [35] proposed a two-level parallel optimization model wherein the multi-objective function was divided into several parallel individual objectives. The first level was used to independently optimize all individual objectives, which were collectively considered at the second level to generate the final solution using a parallel artificial bee colony algorithm. Wu et al. [36] proposed controllable wings for an underwater glider by establishing a multi-objective optimal scheduling model. The optimized Latin hypercube was used to develop a surrogate model (SM), and the Pareto solution set was generated using the NSGA-II. However, their study did not focus on optimal point selection from the Pareto solution set. Deng et al. [37] established a sandwich corrugated square tube optimization model. To simplify the optimization process, an SM was developed using a radial basis function, and multi-objective particle swarm optimization (MOPSO) was used to generate the optimal solution. Li et al. [38] proposed an optimization model for a desiccant wheel. First, the Pareto solution set was generated using the NSGA-II, and the final optimal point was selected using the technique for order preference by considering the similarity to TOPSIS.

Therefore, the MOSS solution process has three steps: developing an SM, establishing a Pareto frontier set, and choosing the final optimal solution.

The SM is employed to replace the complex actual model and simplify the solution process. Therefore, the SM must demonstrate an excellent feature-extraction ability to accurately map the relationship between the input and output using a few parameters. Compared with other methods, the DNN technology can better complete feature extraction and is more suitable for use as an SM [39]. Among all DNNs, the stacked multilevel denoising autoencoder (SMLDAE) contains multiple noise levels that effectively facilitate complete feature extraction [40].

Mainstream approaches for establishing the Pareto frontier set include the NSGA-II, the fruit fly optimization algorithm [41], and MOPSO. The NSGA-II adopts an elite approach and a fast, non-dominated sorting algorithm to ensure a high solving velocity, ensuring that excellent solutions are not lost. Therefore, the NSGA-II has remarkable global optimization performance and is widely used to generate a Pareto frontier set [42].

Upon establishing the Pareto frontier set, the final optimal solution is chosen. Although crucial, previous studies have not considered this step. MCDM is a traditional approach for choosing the final optimal solution and establishes many criteria based on optimization objectives. Subsequently, the weights of each criterion are determined based on the final solution selected [43]. Many intelligent approaches, such as GAs [44] and TOPSIS, have gained increasing attention. However, similar to other types of decision-making problems, the process of choosing the final optimal solution faces challenges. MCDM and intelligent approaches are physical models. Consequently, a large error occurs during the solution process when physical models are used because the final optimal solution selection process is complex, high-dimensional, and non-convex. Concurrently, the setting of parameters in physical models, such as thresholds and weights, depends on subjective experience, making it difficult to guarantee accuracy. Deep reinforcement learning (DRL), as a statistical model, can solve high-dimensional, non-convex processes, and all parameters can be updated automatically until the solving process is complete [45].

Overall, many studies have focused on MCES, including the forecasting model, system modeling, and MOSS. Although each has their own advantages, most of them focus on one aspect. A comprehensive list of previous works is given in Table 1. The complete system, which contains all its aspects, should be addressed to realize the effective application of MCES.

To the best of our knowledge, this is the first study to present the development of a complete MCES comprising renewable-energy forecasting, MCES modeling, and MOSS.

(1) A hybrid forecasting model, including MMMD, BiLSTM, and S2O, based on the RA is developed. RA continuously adds real-time data and updates input datasets to substantially improve forecasting accuracy. MMMD decomposes input data directly in the time domain, effectively reducing forecasting expenses and errors. BiLSTM reserves long-term memory in both time orders, thereby improving the forecasting performance. The S2O approach is designed to overcome the disadvantages of RA.
(2) A campus MCES model based on wind and solar power is designed. Two renewable energy sources (wind and solar) are utilized to increase the effective utilization time. The problems caused by the strong volatility of renewable energy can be solved through the reasonable supplementation of fossil energy and various energy storage systems, ensuring that the energy supply task is completed while emissions are reduced. Finally, based on the different load types on the demand side, an energy supply sequence is planned to decrease the conversion times of different energy forms during operation and increase the energy utilization rate.
(3) A novel MOSS comprising SMLDAE-NSGA-II and DRL is proposed. An SM is developed using SMLDAE to simplify the solution process. Based on the developed SM, the Pareto frontier set is established using the NSGA-II, and the final optimal solution is chosen using DRL. To the best of our knowledge, this is the first study to use DRL for selecting the final optimal solution.

Fig. 1 illustrates the framework of the proposed campus MCES.

2. Methodology

2.1. Forecasting model

A hybrid forecasting model comprising MMMD, BiLSTM, and S2O based on RA was developed to forecast changes in wind power, PV power, and solar heating. The input data are continuously updated by RA, which adds new data and removes the old data in a rolling manner to increase the forecasting accuracy. Subsequently, the input data are fed into the MMMD to be decomposed into a series of subsequences directly in the time domain, effectively reducing the conversion time and errors. Thereafter, the forecasting of all subsequences is achieved through BiLSTM, which can reserve long-term memories in both time orders, effectively improving the forecasting performance. Finally, the forecasting results of all subsequences are aggregated to obtain the final

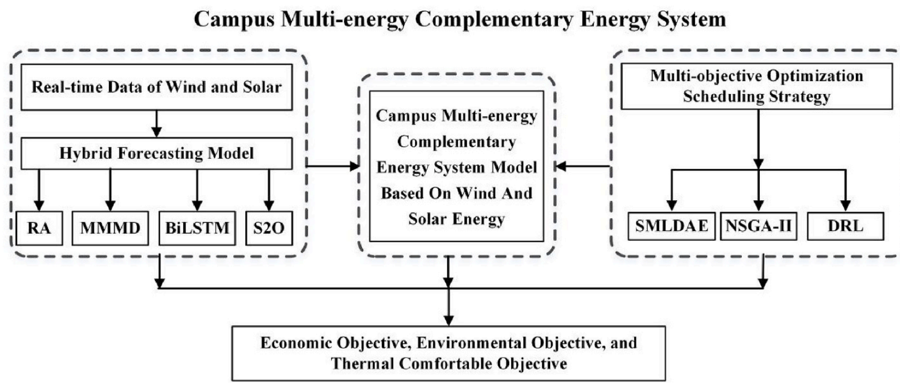

Fig. 1. Framework of the proposed campus MECS.

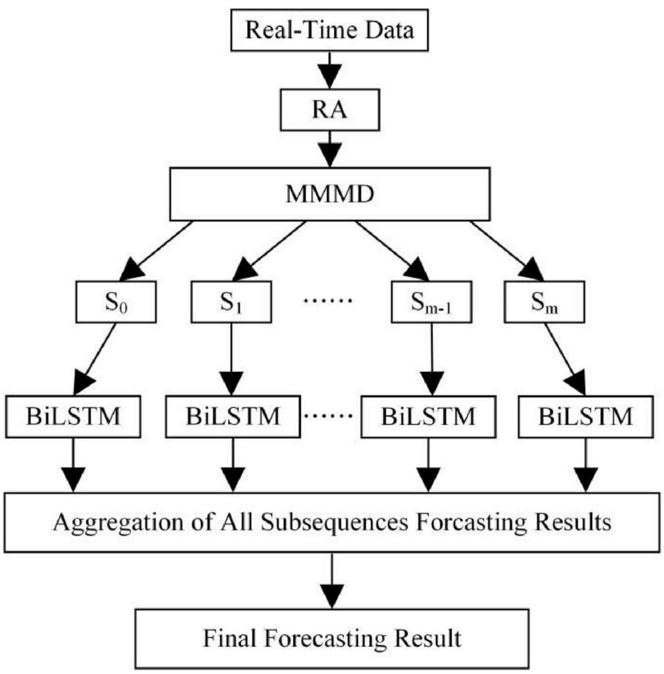

Fig. 2. Schematic of the proposed forecasting model.

forecasting result. Both MMMD and BiLSTM are trained and tested in an S2O manner, which reduces the number of required modules and decreases the forecasting cost. The schematic diagram of the proposed forecasting model is shown in Fig. 2. The detailed modeling process is shown in Supplementary Material A.

2.2. Campus MCES model

Fig. 3 shows the structure of the proposed campus MCES model. In the proposed MCES model, two renewable energy sources (wind and solar) are utilized to increase the effective utilization time. The problems caused by the strong volatility of renewable energy can be solved through the reasonable supplementation of fossil energy and various energy storage systems, ensuring that the energy supply task is completed while emissions are reduced. Based on the different load types on the demand side, an energy supply sequence is planned to decrease the conversion times of different energy forms during operation and increase the energy utilization rate.

The energy inputs include wind energy, solar energy, and natural gas. Energy conversion is enabled by a wind generator (WG), PV array, solar thermal collector (STC), gas-turbine generator (GTG), gas boiler (GB), fuel cell (FC), electric chiller/heater (ECH), absorption chiller/heater (ACH), and heat-recovery boiler (HCB). Energy storage devices include heat storage tanks (HSTs), battery energy storage systems (BESSs), and hydrogen production systems (HPSs). The energy output includes hot water, electricity, cooling, and heating.

The energy hub (EH) model is used for describing energy generation, transmission, conversion, storage, and other coupling relationships. Fig. 4 illustrates the energy flowchart of the proposed campus MCES based on the EH model.

Electrical loads are first supplied by the WG and PV, then by the BESS and FC, and finally by the GTG, powered by natural gas. Excess energy is stored in the BESS when renewable energy exceeds the electrical load demand. The HPS produces hydrogen to store energy once the BESS reaches its rated capacity. When renewable energy is insufficient, the BESS, hydrogen-powered FC, and GTG provide supplements. The heating and hot-water loads were first sacrificed by the STC, followed by HST and ECH powered by electricity from the BESS, and finally by ACH. The HCB collects high-temperature exhaust gas to preheat the reaction of the FC. The cooling load was first sacrificed by the ECH, powered by electricity in the BESS, and then by the ACH. The energy flow of the MCES can be modeled as follows:

$$\begin{aligned}&\widehat{P}_{w}(t)+\widehat{P}_{pv}(t)+P_{GTG}(t)+\widehat{P}_{STC}(t)+P_{ACH}(t)+P_{ECH}(t)+P_{HY}(t)+P_{BESS}(t)\\&\quad+\quad P_{HST}(t)\\&\quad=P_{LE}(t)+P_{LC}(t)+P_{LH}(t)+P_{LW}(t),\end{aligned} \tag{1}$$

where $\widehat{P}_{w}(t)$, $\widehat{P}_{pv}(t)$, and $\widehat{P}_{STC}(t)$ represent the forecasting result of wind power, PV power, and heat generated using solar energy at time t, respectively; $P_{GTG}(t)$ represents the electricity generated through GTG at time t; $P_{ACH}(t)$ and $P_{ECH}(t)$ represent the heat generated through ACH and ECH at time t, respectively; $P_{HY}(t)$, $P_{BESS}(t)$, and $P_{HST}(t)$ represent the power or heat flowing from the HPS, BESS (except ECH), and HST at time t, respectively; $P_{LE}(t)$, $P_{LC}(t)$, $P_{Lh}(t)$, and $P_{LW}(t)$ represent power or heat consumption of the electrical, cooling, heating, and hot-water loads at time t, respectively.

Since solar and wind data are time-varying, the wind power, PV power, and heat generated using solar energy at time t use the forecasted values ($\widehat{P}_{w}(t)$, $\widehat{P}_{pv}(t)$, and $\widehat{P}_{STC}(t)$). In this study, DRL is utilized to complete the optimal solution selection process. Moreover, we use the forecasted values ($\widehat{P}_{w}(t)$, $\widehat{P}_{pv}(t)$, and $\widehat{P}_{STC}(t)$) to train the DRL model, enabling us to effectively eliminate the error of the forecasting value.

The natural gas supply is divided into two parts: one part is supplied to the GTG for electricity generation, while the other part is supplied to the GB to drive the ACH for heating or cooling loads. The energy produced by the BESS is divided into two parts: one part is used to balance the power supply when renewable energy generation is extremely high or low ($P_{BESS}(t)$), and the other part is used to drive the ECH to satisfy the

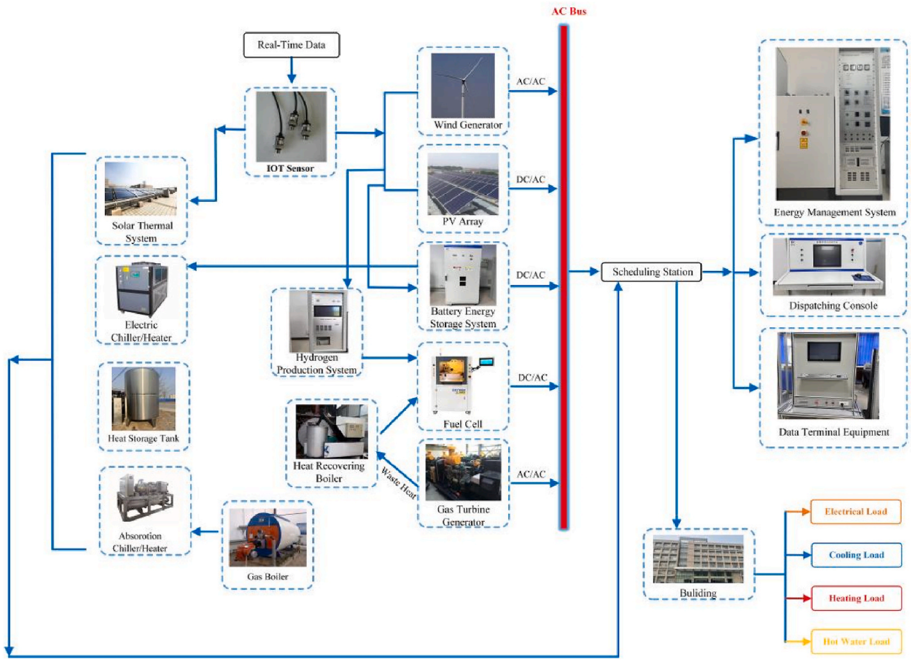

Fig. 3. Structure of the proposed campus MCES model.

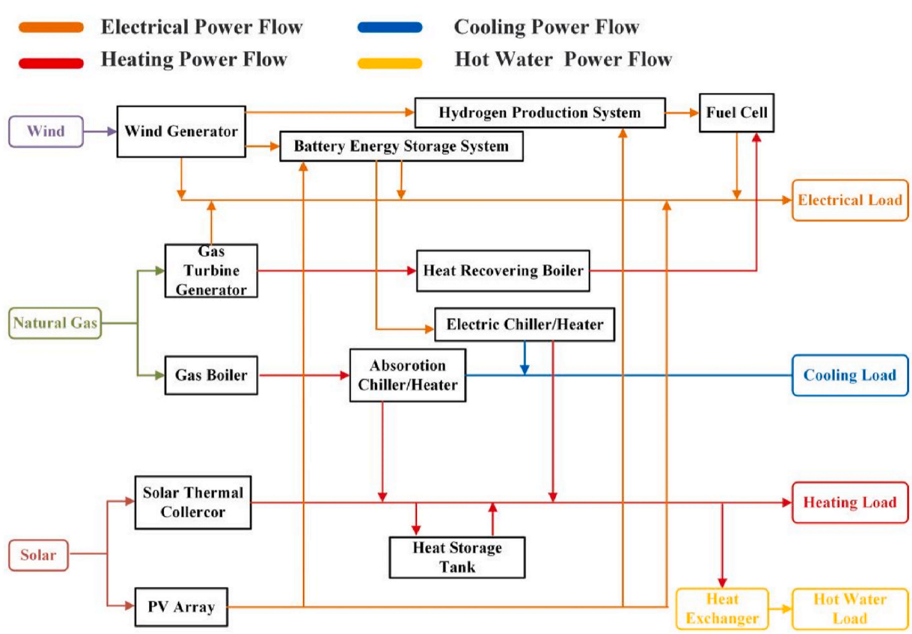

Fig. 4. Energy flowchart of the proposed campus MCES.

demand of the heating load. Once the BESS reaches its rated capacity, the excess electricity is converted to hydrogen for storage. When the power is insufficient, hydrogen is discharged through the fuel cell. Therefore, two possible forms exist for $P_{HY}(t)$: electricity stored as hydrogen and electricity released via the FC.

The detailed modeling steps of MCES are shown in Supplementary material B.

2.3. MOSS

The decision variables are forecast values of wind power ($\widehat{P}_{w}(t)$), PV power ($\widehat{P}_{pv}(t)$), heat from STC ($\widehat{P}_{STC}(t)$), electricity generated through GTG ($P_{GTG}(t)$), heat generated through ACH ($P_{ACH}(t)$) and ECH ($P_{ECH}(t)$), power flowing from hydrogen production (P_{HY}), power or heat flowing from BESS (except ECH) ($P_{BESS}(t)$) and HST ($P_{HST}(t)$), and power or heat consumption of four different loads ($P_{LE}(t)$, $P_{LC}(t)$, $P_{LH}(t)$, and $P_{LW}(t)$). The scheduling objectives include economic, environmental, and thermal comfort. A novel scheduling strategy comprising SMLDAE-NSGA-II, and DRL was proposed to achieve the multi-objective goal.

2.3.1. Optimization objectives

The economic objective is to minimize the cost during the operating process of the campus MCES, while the environmental and thermal comfort objectives are to reduce environmental pollution and daily unsatisfied hours. The MOSS model is expressed as follows:

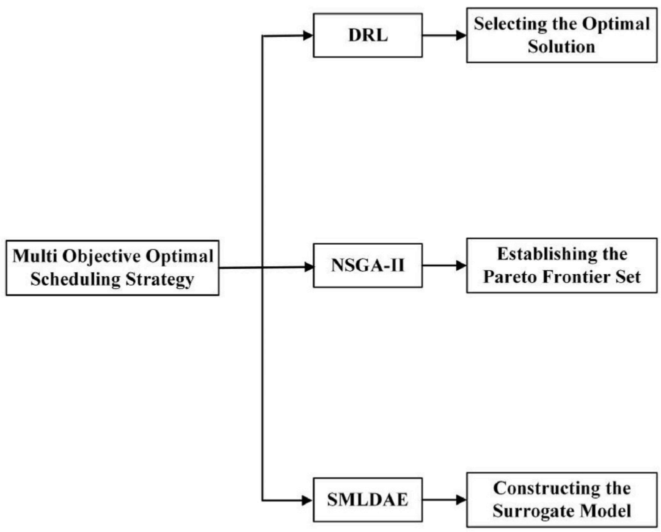

Fig. 5. Overall scheduling flowchart.

$$\begin{cases} \min A(v) = \min\left[A_1(v), A_2(v), A_3(v)\right] \\ S(v) \leq 0 \\ E(v) = 0 \\ v(t) = \left[\widehat{P}_w(t), \widehat{P}_{pv}(t), \widehat{P}_{STC}(t), P_{GTG}(t), P_{ACH}(t), P_{ECH}(t), P_{HY}(t), P_{BESS}(t) \right. \\ \left. P_{HST}(t), P_{LE}(t), P_{LC}(t), P_{LH}(t), P_{LW}(t) \right] \end{cases}, \tag{2}$$

Here, $A(v)$ represents the objective function and $A_1(v), A_2(v)$, and $A_3(v)$ represent the economic, environmental, and thermal comfort objectives, respectively. The economic, environmental, and thermal comfort benefits correspond to objectives $A_1(v), A_2(v)$, and $A_3(v)$. The lower the values of $A_1(v), A_2(v)$, and $A_3(v)$, the higher the economic, environmental, and thermal comfort benefits. All three objectives have the same weight, i.e., one. Moreover, v represents a set of decision variables, $S(x)$ represents the inequality constraint, and $E(x)$ represents the equality constraint. The detailed content of different objectives and constraints is shown in Supplementary Material C.

2.3.2. Optimal scheduling strategy

A novel MOSS comprising SMLDAE-NSGA-II and DRL is presented in this section.

SMLDAE is used to construct the surrogate model, which is the model that simplifies the original complex model. NSGA-II is used to establish the Pareto frontier set, and DRL is used to select the optimal solution. First, historical data are used to train SMLDAE to develop the surrogate model that replaces the original complex model. The noise level, which is a substantial factor in the denoising autoencoder model, affects the robustness of the model and the learning representation. SMLDAE can use multiple noise levels to train a single encoder, which has a stronger feature extraction than other models, enabling it to better complete the function of the surrogate model. During the scheduling process, all decision variables are input into the surrogate model, and then NSGA-II is used to obtain the Pareto frontier set. While obtaining high solving speed, it ensures that excellent solutions are not lost. Finally, DRL is used to choose the optimal solution from the Pareto frontier set as the specific scheduling execution scheme. DRL, as a statistical model, can more effectively solve high-dimensional, non-convex processes, and all parameters can be updated automatically until the end of the solving process. The overall scheduling flowchart is illustrated in Fig. 5.

2.3.2.1. SMLDAE. The first step involves developing an SM to replace the complex actual model using the SMLDAE. The SMLDAE contains multiple noise layers to train the model, which has robust feature extraction abilities [46].

In this study, the SMLDAE was modeled using Python 3.10, and the main hyperparameters were set automatically. Six noise layers ($N_{l0} > N_{l1} > \cdots > N_{l5}$) and five DAE layers ($DAE_1, DAE_2, \cdots, DAE_5$) exist. The training process of the SMLDAE was similar to that of the traditional SDAE; the only difference was that each DAE layer was trained under six noise layers. Specifically, the training process began with DAE_1, which was trained from N_{l0} to N_{l5}. Other DAEs were also trained similarly, and the model parameters and hidden representations were set. The activation function and objective function are the sigmoid function and root mean square error (RMSE) (detailed in Eq. (35)), respectively.

Owing to the limitation in the length of the manuscript, the specific modeling process is described in our previous publications [40,47].

2.3.2.2. NSGA-II. Upon developing SM, the Pareto frontier set was generated based on the decision variables through NSGA-II. The Pareto frontier set was generated using objective functions ($A(v)$) with good mean values. The procedure for the NSGA-II algorithm used in this study is as follows:

(1) All parameters are initialized. The population size (S_p) was 200, evolution generation (E_g) was 500, crossover probability (P_{cross}) was 0.9, mutual probability (P_{mut}) was 0.1, and both mutual distribution (D_{mut}) and crossover distribution (D_{cross}) were 20. The initial population ($N_{initial}$) containing S_p chromosomes was randomly established under all constraints, and the objective functions for each individual were solved based on Equations 1–29. For each individual, decision variables $v = (\widehat{P}_w(t), \widehat{P}_{pv}(t), \widehat{P}_{STC}(t), P_{GTG}(t), P_{ACH}(t), P_{ECH}(t), P_{HY}(t), P_{BESS}(t), P_{HST}(t), P_{LE}(t), P_{LC}(t), P_{LH}(t), P_{LW}(t))$, and objective functions $A(v) = (A_1(v), A_2(v), A_3(v))$. Thus, $N_{initial}$ is expressed as

$$N_{initial} = \begin{bmatrix} v_1 & A_1(v_1) & A_2(v_1) & A_3(v_1) \\ v_2 & A_1(v_2) & A_2(v_2) & A_3(v_2) \\ \cdots & \cdots & \cdots & \cdots \\ v_{S_p} & A_1\left(v_{S_p}\right) & A_2\left(v_{S_p}\right) & A_3\left(v_{S_p}\right) \end{bmatrix}, \tag{3}$$

(2) Each initial individual of $N_{initial}$ is non-dominated sorted (NS) and assigned a rank. A crowding distance calculation (CDC) was then executed for each individual. Another matrix with crowding distance and rank was generated as follows:

$$N_{initialNC} = \begin{bmatrix} v_1{}' & A_1(v_1)' & A_2(v_1)' & A_3(v_1)' \\ v_2{}' & A_1(v_2)' & A_2(v_2)' & A_3(v_2)' \\ \cdots & \cdots & \cdots & \cdots \\ v_{S_p}{}' & A_1\left(v_{S_p}\right)' & A_2\left(v_{S_p}\right)' & A_3\left(v_{S_p}\right)' \end{bmatrix}, \tag{4}$$

(3) For executing the tournament selection process, we randomly select two individuals and reserve the one with a higher ranking and crowding degree. Subsequently, the individuals are reduced to half of $N_{initial}$.

$$N_{initialTSP} = \begin{bmatrix} v_1{}'' & A_1(v_1)'' & A_2(v_1)'' & A_3(v_1)'' \\ v_2{}'' & A_1(v_2)'' & A_2(v_2)'' & A_3(v_2)'' \\ \cdots & \cdots & \cdots & \cdots \\ v_{\frac{S_p}{2}}{}'' & A_1\left(v_{\frac{S_p}{2}}\right)'' & A_2\left(v_{\frac{S_p}{2}}\right)'' & A_3\left(v_{\frac{S_p}{2}}\right)'' \end{bmatrix}, \tag{5}$$

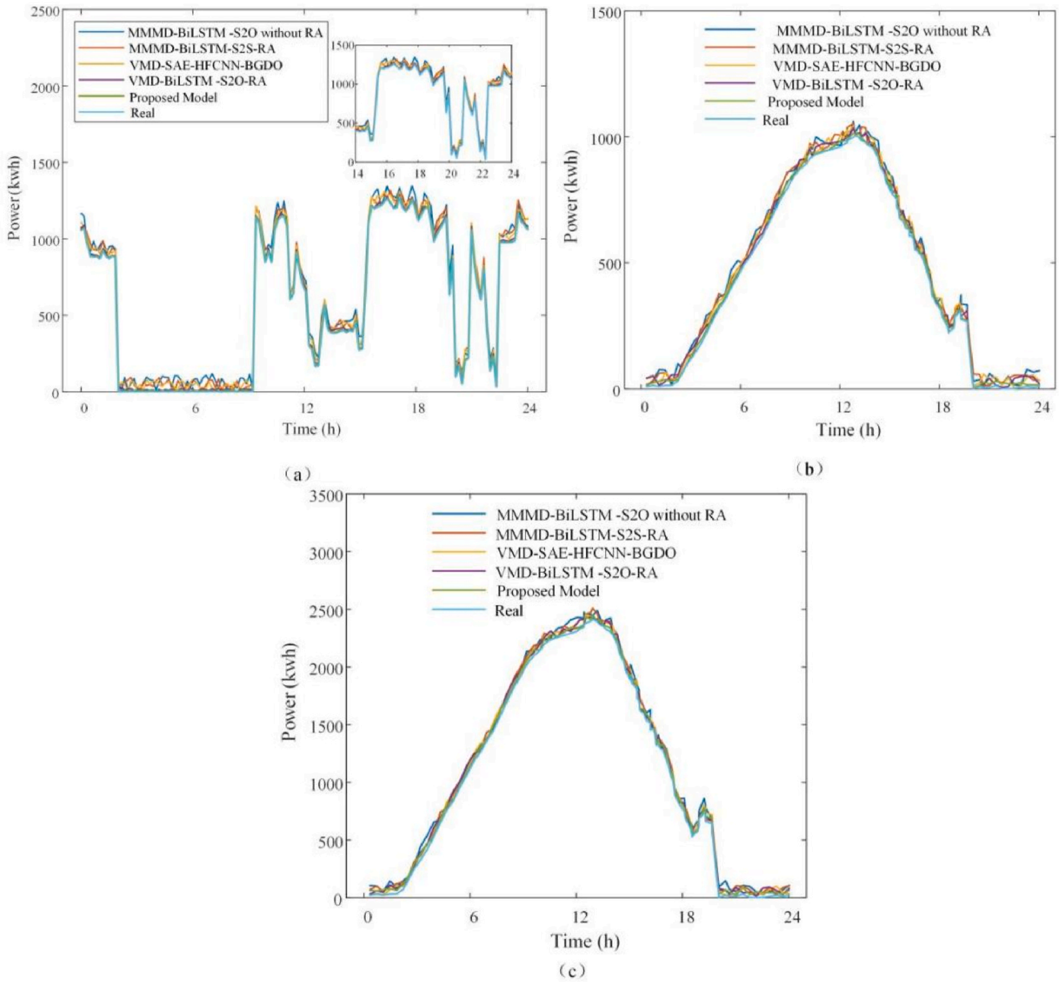

Fig. 6. Forecasting results of (a) wind power, (b) PV power, and (c) solar heating.

Table 2
Comparison of RMSE and MAE.

Model	Wind/PV/STC	RMSE	MAE
Proposed model	Wind power	**8.12**	**0.1739**
	Solar heating	**7.97**	**0.1679**
	PV power	**7.89**	**0.1676**
VMD-SAE-HFCNN-BGDO	Wind power	9.96	0.1797
	Solar heating	8.37	0.1789
	PV power	8.12	0.1755
VMD-BiLSTM -S2O-RA	Wind power	9.87	0.1789
	Solar heating	8.23	0.1776
	PV power	8.19	0.1766
MMMD-BiLSTM-S2S-RA	Wind power	15.21	0.5378
	Solar heating	15.03	0.5359
	PV power	15.07	0.5361
MMMD-BiLSTM -S2O without RA	Wind power	17.89	0.6012
	Solar heating	17.81	0.6001
	PV power	17.68	0.5987

Table 3
Comparison of different models.

Model	A_1	A_2	A_3
Proposed model	11.8	2.1	2.6
Wind-based model	9.2	3.5	4.1
Fossil energy model	4.8	10	0

(4) The offspring N_{os} is established by a polynomial mutation operation and a genetic operation through binary crossover. Two offsprings are generated by each father generation, and a total of S_p individuals are generated in each offspring generation.

$$N_{os} = \begin{bmatrix} v_{1os} & A_1(v_{1os}) & A_2(v_{1os}) & A_3(v_{1os}) \\ v_{2os} & A_1(v_{2os}) & A_2(v_{2os}) & A_3(v_{2os}) \\ \cdots & \cdots & \cdots & \cdots \\ v_{S_pos} & A_1(v_{S_pos}) & A_2(v_{S_pos}) & A_3(v_{S_pos}) \end{bmatrix}, \tag{6}$$

(5)$N_{initial}$ and N_{os} are combined to generate a new 2 S_p population N_{snew}. NS and CDC are then executed for N_{snew}.

(6) A new population $N_{initial+1}$ is established by choosing S_p individuals from N_{snew} with a higher ranking and crowding degree.

(7) The iteration in this study was set to 10000. After the iteration ended, the Pareto solution set was obtained.

2.3.2.3. DRL. To select the optimal solution in the Pareto solution set, the DRL was established, which included five elements: agent, environment, state, action, and reward. The agent first observes the current state of the environment and then performs the necessary actions based on the demand. The environment changes to the next state and provides reward feedback to the agent. The agent constantly adjusts the action based on the feedback reward until the best effect is achieved. In this study, the environment comprised all the devices and loads in the MCES, and the agent was the scheduling station. The state space comprises the

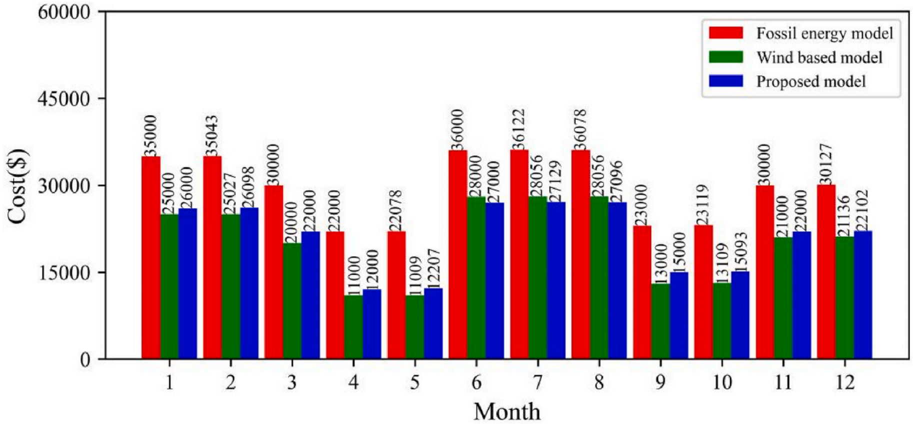

Fig. 7. Monthly average cost (excluding investment cost) for 2021.

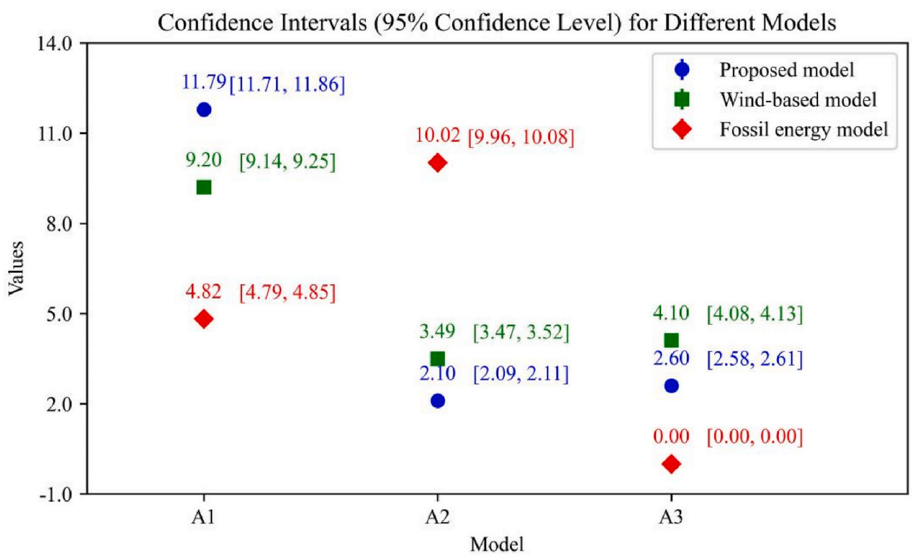

Fig. 8. Confidence intervals of different models.

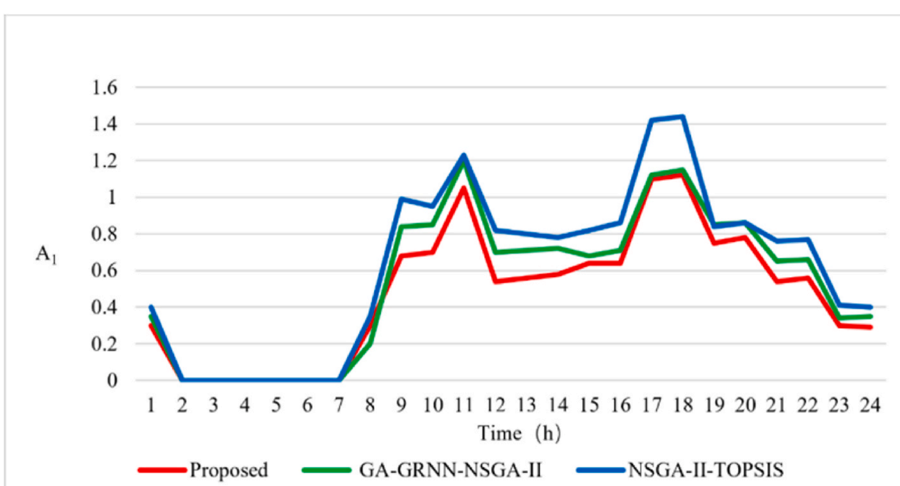

Fig. 9. Economic comparison of different strategies.

forecasting results of $P_w(t), P_{pv}(t)$, and $P_{STC}(t)$; output value of GTG and ACH; the remaining electricity, heat, or hydrogen of the BESS, HST, and hydrogen storage tank; and the demand for four different loads. The action space comprises all the solutions in the Pareto solution set. The DRL was solved using a deep Q-network. The agent predicts the execution probability of each action and chooses the one with the

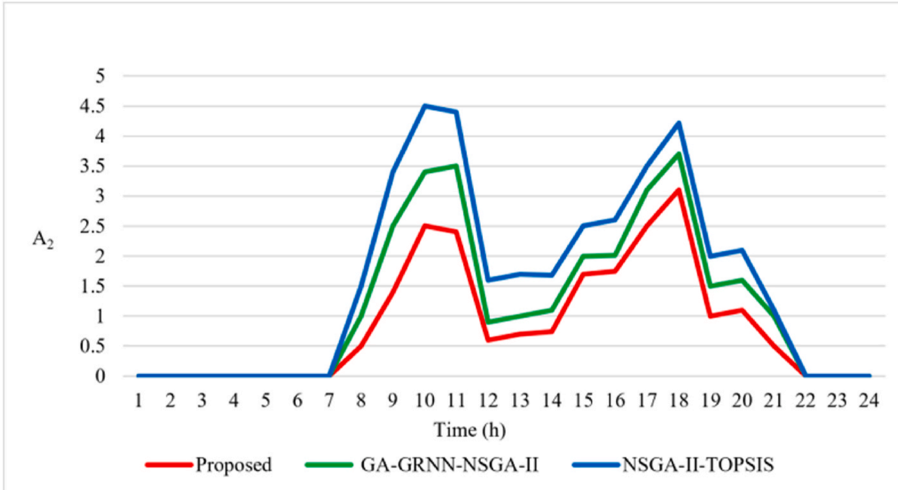

Fig. 10. Environmental comparison of different strategies.

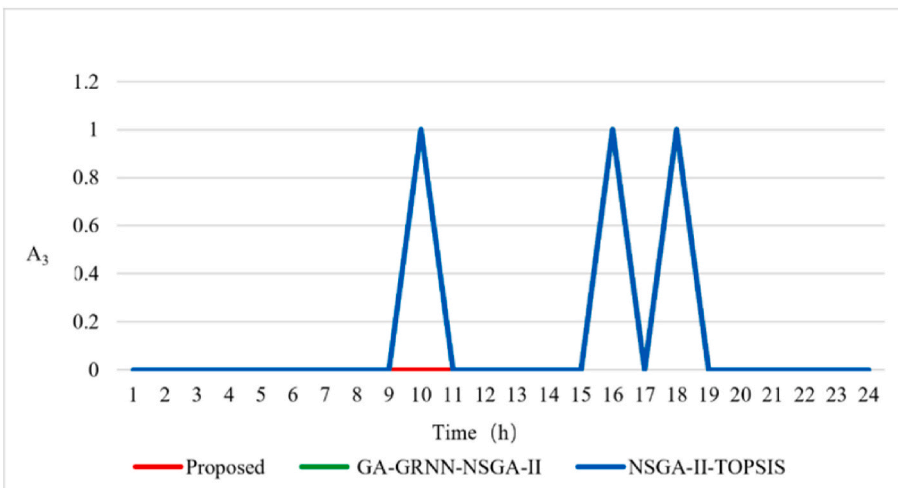

Fig. 11. Thermal comfort comparison of different strategies.

highest probability, obtaining reward R.

$$R = X - A_1(v) - A_2(v) - A_3(v), \tag{7}$$

Here, X is a constant. A larger reward is obtained when A(v) is smaller.

3. Case study

3.1. Test of proposed forecasting model

All the real-time data (e.g., wind and solar data, power or heat generated from different equipment, and power or heat consumption of different loads) were collected at 15-min observation intervals using Internet of Things (IoT) sensors. The data were collected from 2021/1/1 to 2023/12/31. Two commonly used criteria, the RMSE and mean absolute error (MAE), were used to evaluate the forecasting results [20,26,47].

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{A}\left[P(t) - \widehat{P}(t)\right]^2}{A}}, \tag{8}$$

$$\mathrm{MAE} = \frac{\sum_{i=0}^{A} P(t) - \widehat{P}(t)}{A}, \tag{9}$$

where $\widehat{P}(t)$ and $P(t)$ represent the forecast and actual values at time t, respectively, and A indicates the total number of samples.

Three benchmark models, such as VMD-BiLSTM-S2O-RA, MMMD-BiLSTM-S2O without RA, and MMMD-BiLSTM-subsequences to subsequences (S2S)-RA, along with the latest hybrid model, VMD-SAE-HFCNN-BGDO [17], were used to verify the performance of the proposed model (MMMD-BiLSTM-S2O based on RA).

Fig. 6 presents the forecasting results of the different modes on a randomly selected day, and Table 2 presents the comparison results of the RMSE and MAE.

The only difference between the proposed model and VMD-BiLSTM-S2O-RA is the decomposition tool. Fig. 6 and Table 2 suggest that, as a decomposition tool, MMMD yields better results than VMD. However, only a small gap exists between the two models. MMMD-BiLSTM-S2S-RA and MMMD-BiLSTM-S2O without RA were used to demonstrate the superiority of S2O and RA, respectively. The comparison results also indicate that both S2O and RA improve the accuracy of the forecasting model relative to S2S and without RA.

In the latest hybrid model, the performance of VMD-SAE-HFCNN-

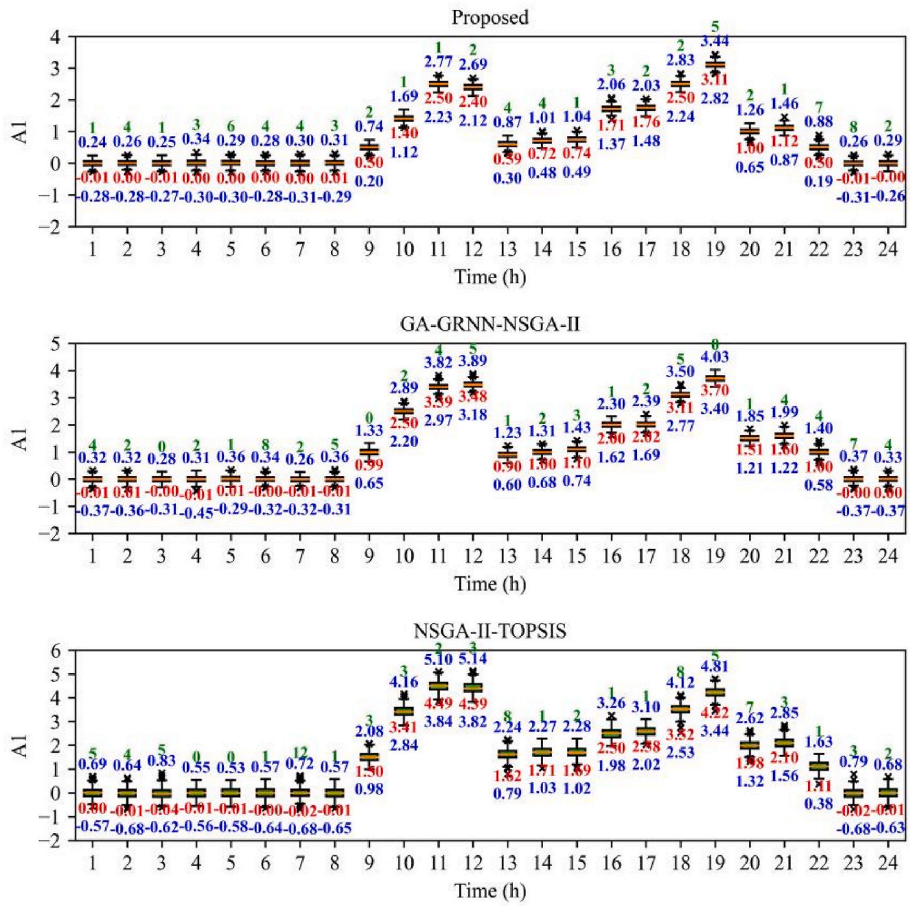

Fig. 12. Box plots of A1 for different strategies.

BGDO was worse than that of the benchmark model, VMD-BiLSTM-S2O-RA. Thus, compared with the traditional hybrid model, which only adopts the accumulation of multi-layer models, the S2O and RA methods can significantly improve the forecasting accuracy.

3.2. Test of the proposed campus MCES model

Owing to limited funds and equipment, the campus MCES developed in this study includes only three teaching buildings and a scheduling station. Each building is equipped with a WG, PV panels, and BESS for power generation. Only one set of other equipment, including STC, GTG, GB FC, HPS, HCB, ACH, HST, and ECH, was established to save costs and consider the low-heat demand, which is shared by all buildings. Tables S1 and S2 (Supplementary Material D) present the economic and technical parameters of the equipment, respectively. The selection of the parameters and configuration of the equipment is based on the specific planning requirements of the project, such as the size of the land and capital investment. To simplify the calculation process, we assumed that the lifetime, annual interest rate, and operating loss coefficient of all devices are the same.

In 2021, energy was supplied to the entire campus using fossil fuels (fossil energy model). In 2022, the campus incorporated wind power generation and energy storage systems to reduce the consumption of fossil energy (a wind-based model). Currently, we have constructed MCES models wherein renewable energies and fossil fuels complement each other. Three different models were compared using the same scheduling policy (SMLDAE-NSGA-II-DRL) under three optimization objectives to verify the validity of the proposed model. All models were compared using the 2021 full-year data. A comparison of the results is presented in Table 3 and Fig. 7.

Table 3 lists the results of a randomly selected day for the three objectives. Fossil energy models directly utilize fossil energy and do not require the purchase of new renewable energy equipment. Hence, these models can achieve the best economic value compared with the other two models that incur high costs for purchasing equipment; thus, the value of A_1 is the lowest. Meanwhile, because only fossil energy is used for energy supply, the operation process is stable, and the thermal comfort objective is 100 % guaranteed; hence, A_3 is zero. By contrast, A_2 has the highest value because the fossil energy model did not use renewable energy. Compared with the wind-based model, the proposed model requires purchasing additional equipment; hence, the value of A_1 is higher. However, the proposed model uses two complementary renewable energy operation modes. Thus, the renewable energy access rate is higher, and the value of A_2 is lower. Moreover, the proposed model uses renewable energy complementary to fossil fuels, thereby enhancing stability; thus, A_3 is improved.

Optimal scheduling aims to minimize the value of $A(A_1 + A_2 + A_3)$. The fossil energy model obtained the lowest A value due to its significant advantages in terms of economic cost. However, the inability to use renewable energy makes it impossible to solve environmental problems. Compared to only using wind energy as a renewable energy source, the MCES developed in this study achieved better scheduling results.

Fig. 7 presents the monthly average costs (excluding investment costs) for 2021. The MCES proposed in this study incurs almost the same monthly average cost as the wind-based model, and the cost is lower than that of the fossil energy model.

Further statistical analysis for the results presented in Table 3 is shown in Fig. 8. In the figure, the confidence interval of different models is demonstrated over one year. The width of the confidence interval of all three models is extremely small under a 95 % confidence level, e.g.,

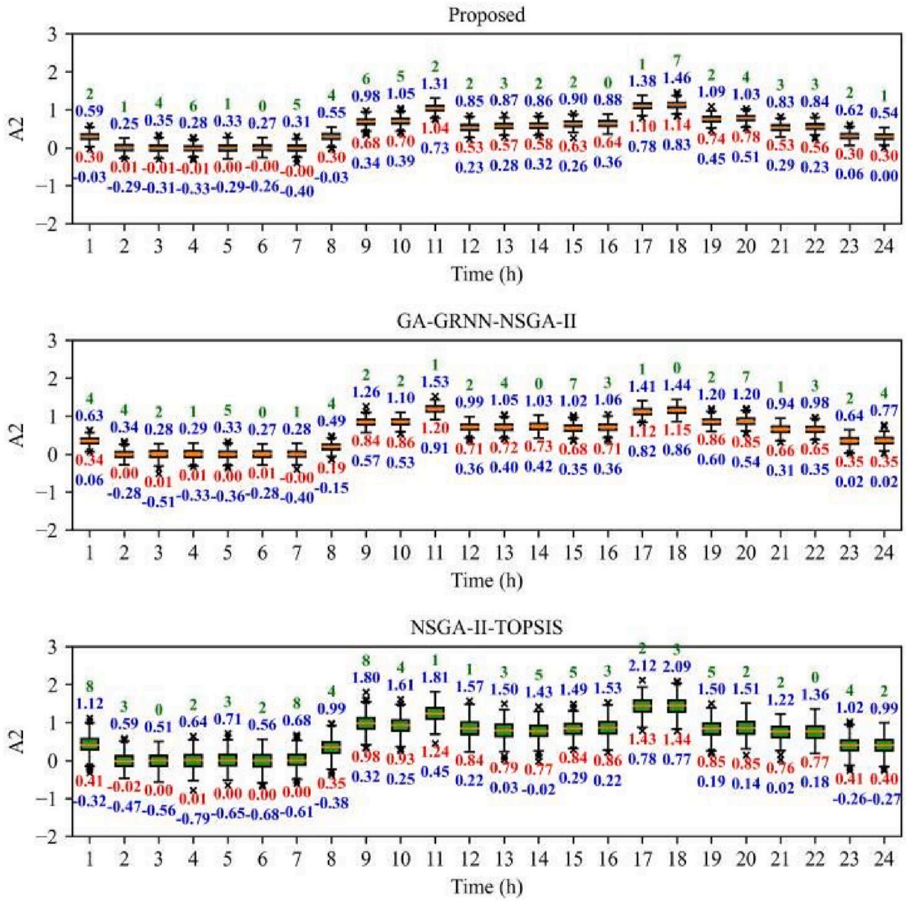

Fig. 13. Box plots of A1 for different strategies.

the width of the confidence interval of the proposed model in A_1 is only 0.15. It means that the results of all three models have a small fluctuation, even for one year of data. Thus, the results presented in Table 3 have statistical significance.

3.3. Test of the proposed MOSS

Two comparison models, NSGA-II-TOPSIS [31] and the GA-generalized regression neural network-NSGA-II (GA-GRNN-NSGA-II) [48], were used to verify the superiority of the proposed scheduling strategy (SMLDAE-NSGA-II-DRL). For the NSGA-II-TOPSIS model, NSGA-II was utilized to generate the Pareto solution set, and TOPSIS was employed to select the optimal solution. All three strategies are verified based on the proposed MCES model. For the GA-GRNN-NSGA-II model, the GRNN optimized by the GA and the NSGA-II were used to construct the SM and generate the Pareto solution set, respectively. The comparison results are presented in Figs. 9–16 and Table 4.

Figs. 9–11 present the comparison results for the different scheduling strategies for each objective on May 10, 2021 (randomly selected). Fig. 9 presents an economic comparison. The teaching building is open from 7:30 a.m. to 10 p.m.; hence, the energy consumption during that period is high. Specifically, there are two peak energy consumption periods—from 10:00 a.m. to 11:30 a.m. and from 4:00 p.m.–6:00 p.m., and a valley time from 12:00 a.m. to 2:00 a.m. From 8:30 p.m. onward, students begin to leave the teaching buildings, and energy consumption begins to decrease. After 10 p.m., the teaching buildings are closed, and only the WG charges the BESS and hydrogen storage system. Moreover, the operation of all equipment stops at approximately 1 a.m. It is evident that the proposed model can obtain the best performance on economic objectives, and the value of A_1 can be reduced by 13.33–29.62 % in comparison with GA-GRNN-NSGA-II and by 10.25–51.85 % compared with NSGA-II-TOPSIS.

Fig. 10 presents the environmental comparison. The peak energy consumption times are 8:30 a.m.–11:30 a.m. and 4:00 p.m.–6:00 p.m. The first peak time is earlier than the economic index, which is primarily because the WG is operated after 9 a.m., and solar energy is low during this time; therefore, fossil energy is required to supply the demand. For the second peak time, the value is high between 5:30 p.m. and 6:00 p.m. because the sun begins to set at 5:30 p.m.; hence, additional fossil fuel energy is required to supply the demand. After 10:00 p.m., the teaching buildings close, and only the WG continues to operate; therefore, the value of the environment index is zero. From the figure, it is clear that the proposed model can provide the best performance in terms of the environmental objective, and the value of A_2 can be reduced by 14.85–100 % in comparison with GA-GRNN-NSGA-II and by 36.12–200 % compared with NSGA-II-TOPSIS.

Fig. 11 shows the thermal comfort comparison. The trend is the same for all three scheduling strategies, so the three curves in the figure mostly coincide. Only during peak times are a few hours available for which the heat or cooling supply cannot satisfy the load demand; therefore, the value of the thermal comfort index is one, while at all other times, its value is zero.

Further statistical analyses of the box plots for objectives A_1, A_2, and A_3 are shown in Figs. 12–14, respectively. Fig. 12 presents the statistical analysis for A_1 of different models for data across the entire year. It is evident that the overall trend of all three models is the same as in Fig. 9, and the range of fluctuations in values throughout the year is extremely small. Only a few outliers exist for all models. The same conclusion can be obtained from Figs. 13 and 14. From Figs. 12–14, it can be concluded that the results presented in Figs. 8–10 are statistically significant,

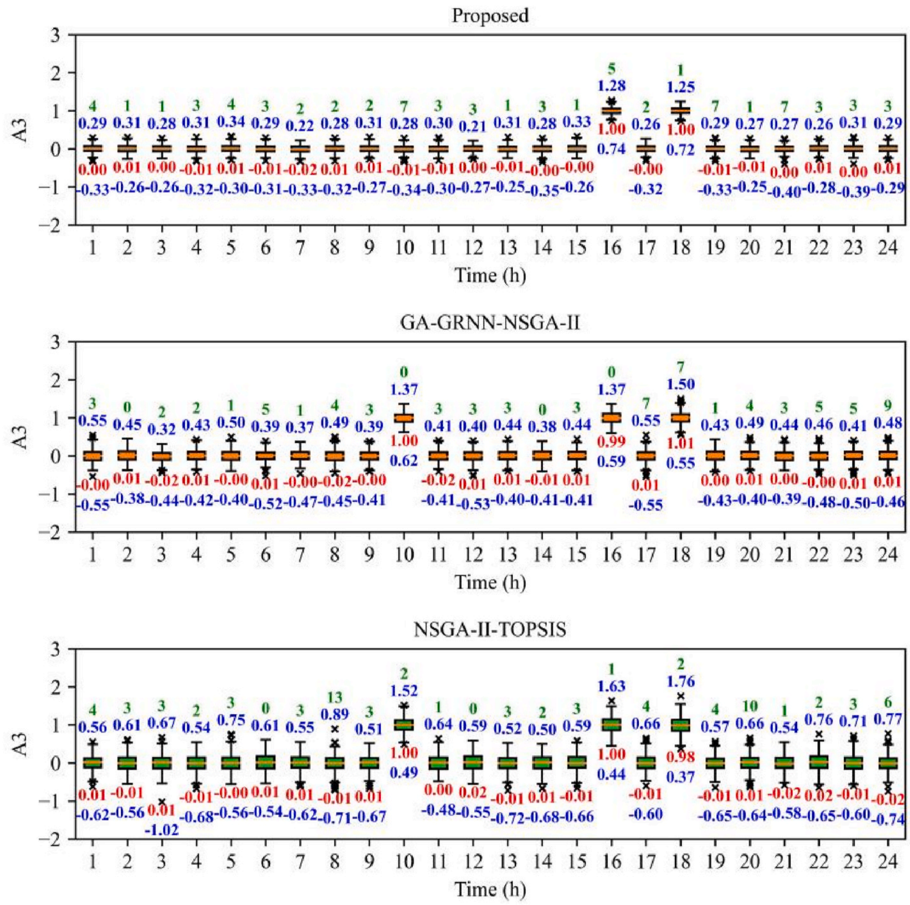

Fig. 14. Box plots of A1 for different strategies.

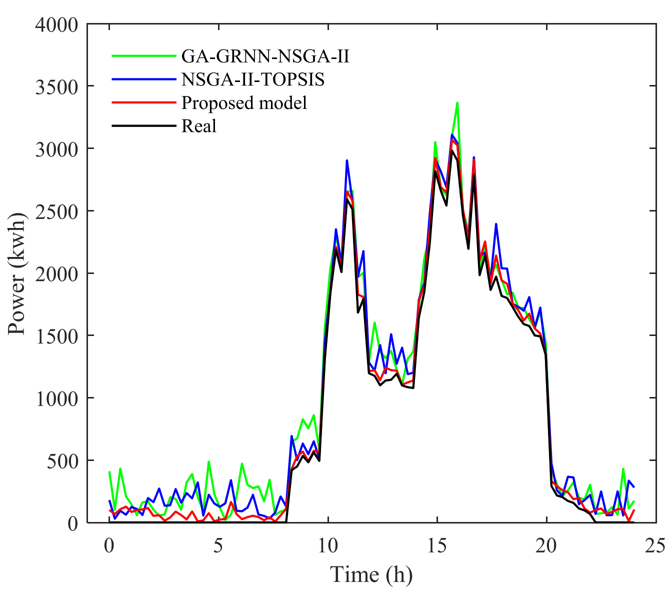

Fig. 15. Comparison of total energy consumption.

whereas no seasonality exists for all the objectives of different models.

Fig. 15 presents a comparison of the total energy consumption on a random day. The energy consumption based on the proposed strategy is closer to the actual value and much smaller than that of the other strategies. Thus, the proposed strategy can better balance the three optimization objectives, achieve reasonable completion of scheduling work, and reduce unnecessary energy losses.

Table 4 lists the comparison results for the daily average values of the different scheduling strategies under the three objectives. The proposed strategy can effectively reduce environmental pollution and costs, but at the expense of the thermal comfort objective (the value of A_3 is larger). However, the total value of $A(A_1 + A_2 + A_3)$ is lower than the corresponding values of the other strategies, indicating that the total reward is high (the calculation of the reward is expressed in Equation (37)). The comparison results of the hourly average reward are shown in Fig. 16, further implying that the proposed strategy can achieve the best reward because the two comparison models have disadvantages.

The comparison results obtained from Figs. 9–16 and Table 4 prove that the proposed model consistently achieves superior scheduling performance compared to other models. The GA-GRNN-NSGA-II model uses only a simple threshold to complete the most important step—-choosing the optimal solution. The approach is simple, and the threshold set is influenced by subjectivity. The NSGA-II-TOPSIS model neglects the SM construction. SM can effectively simplify the solving process and increase accuracy.

4. Conclusion

In this study, a complete MCES comprising an accurate forecasting model, efficient MCES model, and effective scheduling strategy was proposed. The overall power supply capacity of the proposed MCES could independently satisfy the power demand of the entire campus. To the best of our knowledge, this is the first study to design a complete MCES that includes all three parts. The main conclusions of this study are given below.

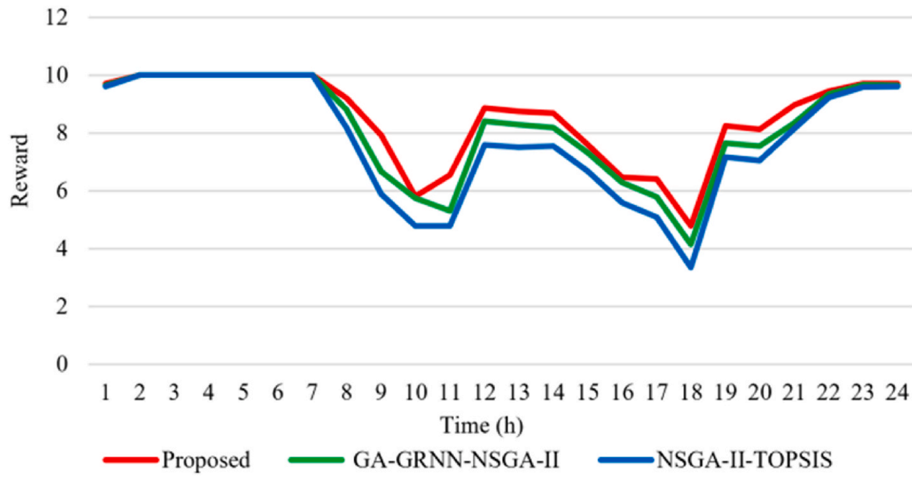

Fig. 16. Hourly average reward of different strategies.

Table 4
Comparison of different scheduling strategies.

Strategy	A_1	A_2	A_3
Proposed	11.9	1.9	2.5
NSGA-II-TOPSIS	12.9	2.5	2.9
GA-GRNN-NSGA-II	13.5	2.7	3.1

(1) A novel hybrid forecasting model, including MMMD, BiLSTM, and S2O, was developed based on the RA. RA was used to update input data continuously. MMMD and BiLSTM were used to decompose the input data and perform forecasting. The S2O method reduced the number of required modules. Compared with other forecasting models, the RMSE and MAE of the proposed model are reduced by 0.23–9.84 and 0.009–0.4322, respectively.
(2) A campus MCES model based on two renewable energy sources was designed. The model could effectively increase the renewable energy utilization time, mitigate the volatility, and increase the energy utilization rate. The proposed MCES better balances the multiple requirements, such as the utilization rate of renewable energy and economic costs.
(3) A novel MOSS containing SMLDAE-NSGA-II-DRL was proposed. Modules SMLDAE, NSGA-II, and DRL were used for SM building, Pareto frontier set establishment, and optimal solution selection, respectively. To the best of our knowledge, this is the first study to use DRL to select the final optimal solution. Compared with other strategies, the daily average values for the three objectives of the proposed strategy increased by 1–1.4, 0.6–0.8, and 0.4–0.6, respectively.

The complete MCE developed in this study is only a preliminary model, and more analysis and discussion are required, e.g., overall computational load calculation and robustness analysis of strategy. Moreover, a more realistic model that can consider the lifespan, annual interest rate, and operational loss coefficient difference of each piece of equipment needs to be developed, and an uncertainty analysis should be proposed. We plan to address these aspects in future research.

Funding

This work was supported by the National Natural Science Foundation of China (51877070 and 51577048), Natural Science Foundation of Hebei Province (E2018208155 and E2018210044), Hebei Education Department Fund, S&T Program of Hebei (20314501D and 19214501D), Science and Technology Project of the Hebei Education Department (ZD2022030), and the Hebei Education Department Fund (BJK2024091).

Data and code availability

Data and code will be made available on reasonable request.

CRediT authorship contribution statement

Weichao Dong: Writing – original draft, Methodology, Conceptualization. **Hexu Sun:** Supervision, Project administration. **Zheng Li:** Funding acquisition. **Huifang Yang:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Glossary

Absorption chiller/heater ACH
Batch gradient descent optimization BGDO
Bidirectional long short-term memory BiLSTM
Crowding distance calculation CDC
Convolutional neural network CNN
Deep neural network DNN
Deep reinforcement learning DRL
Electric chiller/heater ECH
Energy hub EH
Fuel cell FC
Genetic algorithm GA
Gas boiler GB
Gas-turbine generator GTG
Heat-recovery boiler HCB
High-order fuzzy cognitive mapping neural network HFCMNN
Hydrogen production system HPS
Heat storage tank HST
Mean absolute error MAE
Multi-energy complementary energy system MCES
Multi-criteria decision-making MCDM
Multi-scale mathematical morphological decomposition MMMD

Multi-objective particle swarm optimization MOPSO
MOSS, multi-objective optimal scheduling strategy MOSS
Non-dominated sorting genetic algorithm-II NSGA-II
Non-dominated sorted NS
Photovoltaic PV
Rolling approach RA
Root mean square error RMSE
Subsequences to the original sequence S2O
Sparse autoencoder SAE
Stacked multilevel-denoising autoencoder SMLDAE
Surrogate model SM
Solar thermal collector STC
Variable mode decomposition VMD
Wind generator WG
Wavelet transform WT

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2024.133088.

References

[1] Kuriqi A, Pinheiro AN, Sordo-Ward A, Bejarano MD, Garrote L. Ecological impacts of run-of-river hydropower plants—current status and future prospects on the brink of energy transition. Renew Sustain Energy Rev 2021;142:110833. https://doi.org/10.1016/j.rser.2021.110833.
[2] Santos RM, Bakhshoodeh R. Climate change/global warming/climate emergency versus general climate research: comparative bibliometric trends of publications. Heliyon 2021;7:e08219. https://doi.org/10.1016/j.heliyon.2021.e08219.
[3] Alabi TM, Agbajor FD, Yang Z, Lu L, Ogungbile AJ. Strategic potential of multi-energy system towards carbon neutrality: a forward-looking overview. Energy Built Environ 2023;4:689–708. https://doi.org/10.1016/j.enbenv.2022.06.007.
[4] Liu Z, Fan G, Sun D, Wu D, Guo J, Zhang S, et al. A novel distributed energy system combining hybrid energy storage and a multi-objective optimization method for nearly zero-energy communities and buildings. Energy 2022;239:122577. https://doi.org/10.1016/j.energy.2021.122577.
[5] Li CZ, Wang LG, Zhang YM, Yu HY, Wang Z, Li L, et al. A multi-objective planning method for multi-energy complementary distributed energy system: tackling thermal integration and process synergy. J Clean Prod 2023;390:135905. https://doi.org/10.1016/j.jclepro.2023.135905.
[6] Zhang JQ, Zhu L, Wang Y, Sun Y, Yan ZX, Zhou B. Global sensitivity analysis and stochastic optimization of multi-energy complementary distributed energy system considering multiple uncertainties. J Clean Prod 2023;389:136120. https://doi.org/10.1016/j.jclepro.2023.136120.
[7] Yang J, Xu W, Ma K, Chen J, Guo W. Integrated demand-side management for multi-energy system based on noncooperative game and multi-energy pricing. Sustain Energy Grids Netw 2023;34:101047. https://doi.org/10.1016/j.segan.2023.101047.
[8] Feng B, Fu Y, Huang Q, Ma C, Sun Q, Wennersten R. Multi-objective optimization of an integrated energy system with high proportion of renewable energy under multiple uncertainties. Energy Rep 2023;9:695–701. https://doi.org/10.1016/j.egyr.2023.05.194.
[9] Ren HB, Quan ZH, Wang ZM, Wang LC, Jing HR, Zhao YH. Performance simulation and analysis of a multi-energy complementary energy supply system for a novel BIPVT nearly zero energy building. Energy Convers Manag 2023;282:116879. https://doi.org/10.1016/j.enconman.2023.116879.
[10] Wang G, Zhang Z, Lin JQ. Multi-energy complementary power systems based on solar energy: a review. Renew Sustain Energy Rev 2024;199:114464. https://doi.org/10.1016/j.rser.2024.114464.
[11] Lu Q, Liu MZ. A multi-criteria compromise ranking decision-making approach for analysis and evaluation of community-integrated energy service system. Energy 2024:132439. https://doi.org/10.1016/j.energy.2024.132439.
[12] Han ZP, Han W, Ye YY, Sui J. Multi-objective sustainability optimization of a solar-based integrated energy system. Renew Sustain Energy Rev 2024;202:114679. https://doi.org/10.1016/j.rser.2024.114679.
[13] Kong XY, Zhang HX, Zhang DL, Huo XX, Peng C. Hierarchical distributed optimal scheduling decision-making method for district integrated energy system based on analysis target cascade. Elec Power Syst Res 2024;234:110533. https://doi.org/10.1016/j.epsr.2024.110533.
[14] Vikash KS, Rajesh K, Ameena SA, Sujil A. Ehsan HF Learning based short term wind speed forecasting models for smart grid applications: an extensive review and case study. Elec Power Syst Res 2023;222:109502.
[15] Huang C, Karimi HR, Mei P, Yang DG, Shi Q. Evolving long short-term memory neural network for wind speed forecasting. Inf Sci 2023;632:390–410. https://doi.org/10.1016/j.ins.2023.03.031.
[16] Zheng L, Lu WS, Zhou QY. Weather image-based short-term dense wind speed forecast with a ConvLSTM-LSTM deep learning model. Build Environ 2023;239:110446. https://doi.org/10.1016/j.buildenv.2023.110446.
[17] Hu YH, Guo YS, Fu R. A novel wind speed forecasting combined model using variational mode decomposition, sparse auto-encoder and optimized fuzzy cognitive mapping network. Energy 2023;278:127926. https://doi.org/10.1016/j.energy.2023.127926.
[18] Shang ZH, Chen Y, Chen YH, Guo ZY, Yang Y. Decomposition-based wind speed forecasting model using causal convolutional network and attention mechanism. Expert Syst Appl 2023;223:119878. https://doi.org/10.1016/j.eswa.2023.119878.
[19] Ahn EJ, Hur J. A short-term forecasting of wind power outputs using the enhanced wavelet transform and arimax techniques. Renew Energy 2023;212:394–402. https://doi.org/10.1016/j.renene.2023.05.048.
[20] Wang SX, Wang X, Wang SM, Wang D. Bi-directional long short-term memory method based on attention mechanism and rolling update for short-term load forecasting. Int J Electr Power Energy Syst 2019;109:470–9. https://doi.org/10.1016/j.ijepes.2019.02.022.
[21] Chen GH, Tang BR, Zeng XJ, Zhou P, Kang P, Long HY. Short-term wind speed forecasting based on long short-term memory and improved BP neural network. Int J Electr Power Energy Syst 2022;134:107365. https://doi.org/10.1016/j.ijepes.2021.107365.
[22] Hu HL, Wang L, Zhang DB, Ling LW. Rolling decomposition method in fusion with echo state network for wind speed forecasting. Renew Energy 2023;216:119101. https://doi.org/10.1016/j.renene.2023.119101.
[23] Wan YF, Jiang J, Zhen ZJ, Zhu P. Multiscale morphology based SVD noise reduction method. Electron Opt Control 2020;27:21–5. 103969/jissn1671 - 637X.
[24] Chen YS, Yu S, Islam S, Lim CP, Muyeen SM. Decomposition-based wind power forecasting models and their boundary issue: an in-depth review and comprehensive discussion on potential solutions. Energy Rep 2022;8:8805–20. https://doi.org/10.1016/j.egyr.2022.07.005.
[25] Veetil IK, Chowdary DW, Chowdary PN, Sowmya V, Gopalakrishnan EA. An analysis of data leakage and generalizability in MRI based classification of Parkinson's Disease using explainable 2D Convolutional Neural Networks. Digit Signal Process 2024;147:104407. https://doi.org/10.1016/j.dsp.2024.104407.
[26] Sun ZX, Zhao SS, Zhang JX. Short-term wind power forecasting on multiple scales using VMD decomposition, K-means clustering and LSTM principal computing. IEEE Access 2019;7:166917–29. https://doi.org/10.1109/ACCESS.2019.2942040.
[27] Nguyen TP, Yeh CT, Cho MY, Chang CL, Chen MJ. Convolutional neural network bidirectional long short-term memory to online classify the distribution insulator leakage currents. Elec Power Syst Res 2022;208:1–11. https://doi.org/10.1016/j.epsr.2022.107923.
[28] Bisoi R, Dash PK, Mishra SP. Modes decomposition method in fusion with robust random vector functional link network for crude oil price forecasting. Appl Soft Comput 2019;80:475–93. https://doi.org/10.1016/j.asoc.2019.04.026.
[29] Wang QS, Duan LQ, Lu ZY, Zheng N. Thermodynamic and economic analysis of a multi-energy complementary distributed CCHP system coupled with solar thermochemistry and active energy storage regulation process. Energy Convers Manag 2023;292:117429. https://doi.org/10.1016/j.enconman.2023.117429.
[30] Liu CR, Wang HQ, Wang ZY, Liu ZQ, Tang YF, Yang S. Research on life cycle low carbon optimization method of multi-energy complementary distributed energy system: a review. J Clean Prod 2022;336:130380. https://doi.org/10.1016/j.jclepro.2022.130380.
[31] Guan TT, Lin HY, Sun Q, Wennersten R. Optimal configuration and operation of multi-energy complementary distributed energy systems. Energy Proc 2018;152:77–82. https://doi.org/10.1016/j.egypro.2018.09.062.
[32] Agency IE. World Energy Outlook. 2023.
[33] Kakavand A, Sayadi S, Tsatsaronis G, Behbahaninia A. Techno-economic assessment of green hydrogen and ammonia production from wind and solar energy in Iran. Int J Hydrogen Energy 2023;48:14170–91. https://doi.org/10.1016/j.ijhydene.2022.12.285.
[34] Chung JY, Park MH, Hong SH, Baek J, Han C, Lee S, et al. Comparative performance evaluation of multi-objective optimized desiccant wheels coated with MIL-100 (Fe) and silica gel composite. Energy 2023;283:128567. https://doi.org/10.1016/j.energy.2023.128567.
[35] Bai YY, Zhang CS, Bai WT. A two-level parallel decomposition-based artificial bee colony method for dynamic multi-objective optimization problems. Appl Soft Comput 2023;147:110741. https://doi.org/10.1016/j.asoc.2023.110741.
[36] Wu QJ, Wu HY, Jiang ZH, Tan LJ, Yang YQ, Yan SZ. Multi-objective optimization and driving mechanism design for controllable wings of underwater gliders. Ocean Eng 2023;286:11553. https://doi.org/10.1016/j.oceaneng.2023.115534.
[37] Deng XL, Yang FM, Cao LB, Huang JL. Multi-objective optimization for a novel sandwich corrugated square tubes. Alex Eng J 2023;74:611–26. https://doi.org/10.1016/j.aej.2023.05.069.
[38] Li HY, Chen YR, Tsai MJ, Huang TF, Chen CL, Yang SF. Multi-objective optimization of desiccant wheel via analytical model and genetic algorithm. Appl Therm Eng 2023;228:120411. https://doi.org/10.1016/j.applthermaleng.2023.120411.
[39] Liang YH, Gou HS, Feng FJ, Liu GS, Huang H. Natural image matting based on surrogate model. Appl Soft Comput 2023;143:110407. https://doi.org/10.1016/j.asoc.2023.110407.
[40] Dong WC, Sun HX, Mei CX, Li Z, Zhang JX, Yang HF, Ding YN. Stochastic optimal scheduling strategy for a campus-isolated microgrid energy management system considering dependencies. Energy Convers Manag 2023;292:117341. https://doi.org/10.1016/j.enconman.2023.117341.
[41] Yang M, Li MT, Yu XG, Huang L, Zheng HY, Miao JY, et al. A performance evaluation method based on the Pareto frontier for enhanced microchannel heat sinks. Appl Therm Eng 2022;212:118550. https://doi.org/10.1016/j.applthermaleng.2022.118550.

[42] Verma S, Pant M, Snasel V. A comprehensive review on NSGA-II for multi-objective combinatorial optimization problems. IEEE Access 2021;9:57757–91. https://doi.org/10.1109/ACCESS.2021.3070634.

[43] Najafi A, Nemati A, Ashrafzadeh M, Hashemkhani Zolfani SH. Multiple-criteria decision making, feature selection, and deep learning: a golden triangle for heart disease identification. Eng Appl Artif Intell 2023;125:106662. https://doi.org/10.1016/j.engappai.2023.106662.

[44] Agarwal G, Gupta S, Ahuja R, Rai AK. Multiprocessor task scheduling using multi-objective hybrid genetic algorithm in fog–cloud computing. Knowl Base Syst 2023;272:110563. https://doi.org/10.1016/j.knosys.2023.110563.

[45] Pateria S, Subagdja B, Tan AH, Quek C. End-to-end hierarchical reinforcement learning with integrated subgoal discovery. IEEE Transact Neural Networks Learn Syst 2022;33:7778–90. https://doi.org/10.1109/TNNLS.2021.3087733.

[46] Jiang GQ, He HHB, Xie P, Tang YF. Stacked multilevel-denoising autoencoders: a new representation learning approach for wind turbine gearbox fault diagnosis. IEEE Trans Instrum Meas 2017;66:2391–402. https://doi.org/10.1109/TIM.2017.2698738.

[47] Dong WC, Sun HX, Tan JX, Li Z, Zhang JX, Zhao YY. Short-term regional wind power forecasting for small datasets with input data correction, hybrid neural network, and error analysis. Energy Rep 2021;7:7675–92. https://doi.org/10.1016/j.egyr.2021.11.021.

[48] Wang ZY, Mulyanto JA, Zheng CR, Wu Y. Research on a surrogate model updating-based efficient multi-objective optimization framework for supertall buildings. J Build Eng 2023;72:106702. https://doi.org/10.1016/j.jobe.2023.106702.